

Business Process Model and Notation (BPMN)

Exam Notes

1 What is BPMN?

BPMN (Business Process Model and Notation) is a **standardized graphical notation** for modeling business processes. It shows the **flow of activities, decisions, and events** in a process so that both business people and technical developers can understand it.

- Maintained by **OMG (Object Management Group)**
- Goal: bridge the gap between **process design** and **process implementation**
- Current common version: **BPMN 2.0**

Exam one-liner: “BPMN is a flowcharting notation for modeling the steps and decision logic of a business process in a standardized, universally understood way.”

2 Core Categories of BPMN Elements

BPMN has **4 basic categories** of elements:

Category	Purpose
Flow Objects	Events, Activities, Gateways — the main building blocks
Connecting Objects	Sequence Flow, Message Flow, Association — link elements together
Swimlanes	Pools & Lanes — show who performs what
Artifacts	Data Objects, Groups, Annotations — extra info, don't affect flow

3 Flow Objects (the “verbs and nouns” of the diagram)

3.1 Events — Circles

Represent something that **happens** during the process.



Start Event
thin circle



Intermediate Event
double-lined circle



End Event
thick/bold circle

Type	Symbol	Meaning
Start Event	Thin circle	Triggers the process (e.g., “Order Received”)
Intermediate Event	Double-lined circle	Happens mid-process (e.g., “Payment Received”)
End Event	Thick/bold circle	Process finishes (e.g., “Order Closed”)

Events can be further typed: **Message**, **Timer**, **Error**, **Signal**, **Escalation** — shown by a small icon inside the circle (envelope = message, clock = timer, etc.)

3.2 Activities — Rounded Rectangles

Represent **work being done**.

- **Task**: a single, atomic unit of work (e.g., “Verify Payment”)
- **Sub-Process**: a compound activity that can be expanded into its own set of sub-steps (shown with a small + icon)

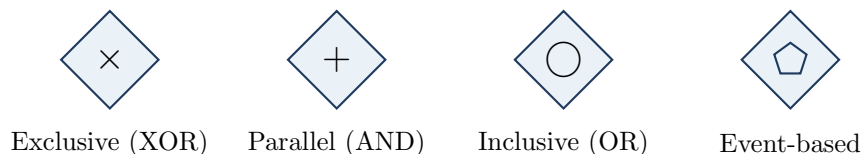


Task types (small icon in top-left corner of the box):

- **User Task** (person icon) — done manually by a person
- **Service Task** (gear icon) — done automatically by a system
- **Script Task** — automated script execution
- **Manual Task** — physical work, no system involved

3.3 Gateways — Diamonds

Represent **decision points / branching or merging logic**.



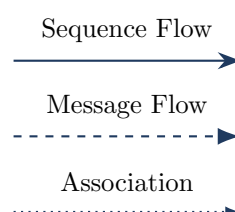
Gateway	Symbol	Meaning
Exclusive (XOR)	Diamond with X or empty	Only ONE path is taken (if/else)
Parallel (AND)	Diamond with +	ALL paths are taken simultaneously
Inclusive (OR)	Diamond with O	ONE OR MORE paths can be taken
Event-based	Diamond with pentagon	Path chosen based on which event occurs first

Exam tip

Examiners love asking “difference between Exclusive and Parallel gateway”:

- Exclusive = choose ONE branch (mutually exclusive conditions)
- Parallel = do ALL branches at once (no condition, just splits work)

4 Connecting Objects



Connector	Line Style	Meaning
Sequence Flow	Solid line, solid arrow-head	Order of activities <i>within</i> one pool
Message Flow	Dashed line, open arrow-head	Communication <i>between</i> two different pools/participants
Association	Dotted line, no/thin arrow-head	Links an Artifact (data object, note) to a flow element

Exam tip

Sequence flow = *within* one participant. Message flow = *across* participants (crosses the pool boundary). This distinction is a favorite exam question.

5 Swimlanes — Who does what

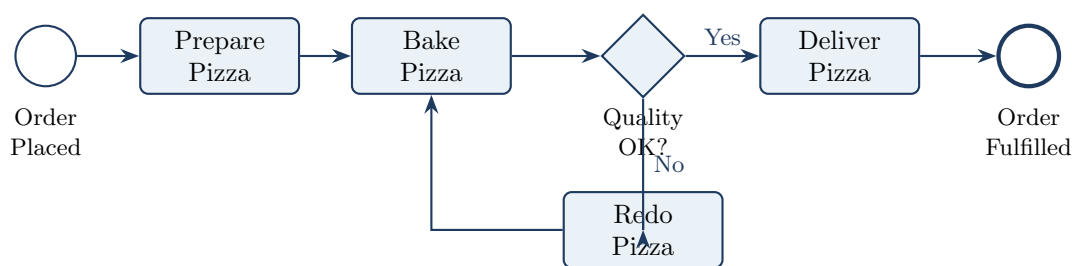
- **Pool:** represents a **participant** (a company, department, or system) — the outer rectangle/container
- **Lane:** a **sub-division inside a pool**, representing a role or sub-department (e.g., “Sales”, “Finance”)

Use swimlanes to show **responsibility** — which actor performs which activity.

6 Artifacts (non-flow-affecting extras)

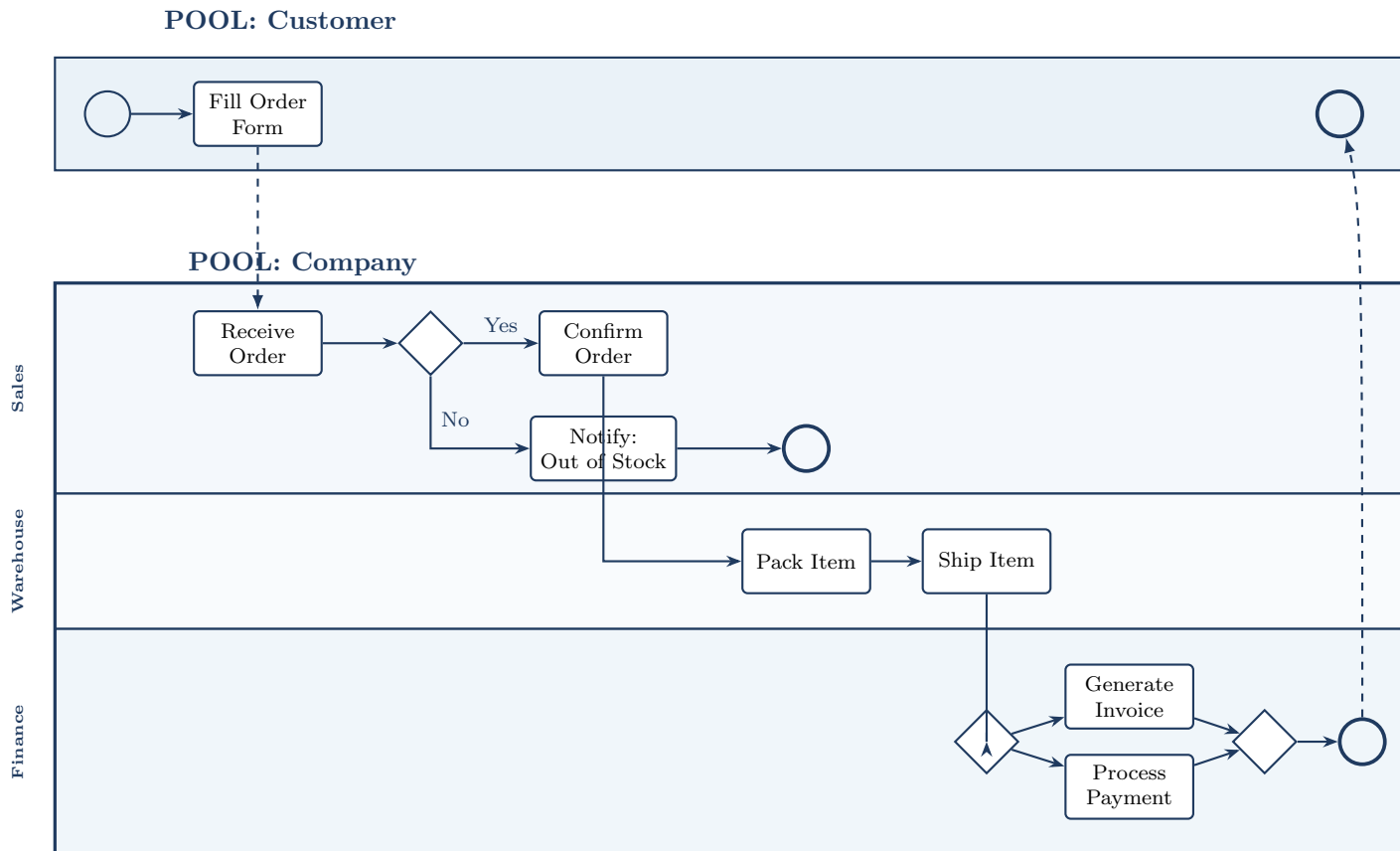
- **Data Object:** represents data used/produced by an activity (rectangle with folded corner)
- **Data Store:** represents a database/persistent data
- **Group:** dotted rounded rectangle grouping related elements (visual only)
- **Text Annotation:** a note attached via association, adds explanation

7 Simple Example Diagram — Pizza Order Process



- Start Event: “Order Placed” (circle)
- Tasks: Prepare, Bake, Deliver, Redo (rounded rectangles)
- Gateway: “Quality OK?” is an **Exclusive Gateway** (only one path taken)
- End Event: “Order Fulfilled” (bold circle)

8 Full Scenario Example with Swimlanes — Online Purchase Order



Textual walkthrough:

- **POOL: Customer** — (Start: Places Order) → [Fill Order Form] $\xrightarrow{\text{Message Flow}}$ enters Company pool
- **POOL: Company, Lane: Sales** — [Receive Order] → <Gateway XOR: In Stock?> Yes → [Confirm Order]; No → [Notify Customer – Out of Stock] → (End: Order Cancelled)
- **Lane: Warehouse** — [Confirm Order] → [Pack Item] → [Ship Item]
- **Lane: Finance** — <Gateway Parallel AND> splits into [Generate Invoice] and [Process Payment], both merge at <Gateway Parallel AND> → (End: Order Completed) $\xrightarrow{\text{Message Flow}}$ back to Customer pool: (End: Order Received)

What this tests in an exam:

- Two **Pools** (Customer, Company) = two separate organizations
- **Message Flow** (dashed) crosses between pools
- **Lanes** inside Company pool show internal role division (Sales, Warehouse, Finance)
- **Exclusive Gateway** used for stock check (only one branch)
- **Parallel Gateway** used for invoice + payment (both happen together, and BOTH must complete before merging — sync point)

9 Common Exam Questions & Quick Answers

Q: What's the difference between a Task and a Sub-Process?

A: A Task is atomic (cannot be broken down further in the diagram). A Sub-Process can be expanded to show its own internal flow of activities.

Q: When do you use a Pool vs a Lane?

A: Pool = separate participant/organization (different company or system). Lane = role/department *within* the same participant.

Q: Why can't Sequence Flow cross a Pool boundary?

A: Because Sequence Flow represents internal control flow within one participant's process. Communication between participants must be shown as Message Flow.

Q: Give an example of when to use an Inclusive Gateway.

A: A customer selects payment methods: they may pay by "Credit Card" AND/OR "Gift Voucher" — one, the other, or both are valid → Inclusive (OR) Gateway, because it's not strictly one path (Exclusive) or all paths (Parallel).

Q: What symbol distinguishes a Start, Intermediate, and End event?

A: Start = thin single-line circle. Intermediate = double-lined circle. End = thick/bold single-line circle.

Q: What is the purpose of BPMN in software engineering (SE) context?

A: Used in the requirements/analysis phase to model business workflows before translating them into system use cases or software specifications — bridges business analysts and developers.

10 Quick Revision Cheat-Sheet

- **Circle** = Event (thin=start, double=intermediate, thick=end)
- **Rounded rectangle** = Activity/Task
- **Diamond** = Gateway (X=exclusive, +=parallel, O=inclusive)
- **Solid arrow** = Sequence Flow (within a pool)
- **Dashed arrow** = Message Flow (between pools)
- **Dotted line** = Association (links artifacts/notes)
- **Big rectangle container** = Pool (participant)
- **Rows inside pool** = Lanes (roles)

Mnemonic: *"Events tell WHEN, Activities tell WHAT, Gateways tell WHICH WAY, Flows tell HOW CONNECTED, Pools/Lanes tell WHO."*